Whose Gold? Annotator-Pool Disagreement Is Large at the Item Level and Invisible in a Six-Model Leaderboard

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Abstract

Preference benchmarks are built by hiring annotators, and the identity of those annotators is treated as an implementation detail. We measure what that detail buys, on two corpora that put *predefined* annotator populations on the *same* items. On the 2,885 MULTIPREF items where both pools are internally unanimous, so no tie-breaking convention is consulted at all, the normal and expert-designated pools assign a different majority label to 23.6% and name the *opposite winner* on 9.2%; on the 246 comparably unanimous MT-BENCH cells, benchmark authors and recruited experts differ on 30.5% and reverse on 8.5%. Yet on both corpora the resulting six-model *orderings* are identical: Kendall $\tau = 1.00$, zero models displaced, even though the win-rate vectors are not.

That invariance is weaker evidence than it looks, and the weakness is measurable on this corpus rather than extrapolated. An item-level bootstrap of the real corpus displaces at least one model in 27.9% of resamples on the full corpus and 34.7% on the convention-free subset the headline uses, almost all of it at the one adjacent pair sitting 0.8pp apart, which the two golds order differently in 22.6% of resamples. These are resampling frequencies on this corpus, not a general probability of leaderboard change. The observed zero is the modal outcome rather than a reliable one, not a property of aggregation, and it is a statement about this leaderboard's spacing rather than about leaderboards. The per-item consumer we test, three local LLM judges, leans toward the normal pool by 1.8–3.5pp after reweighting to the corpus when scored against tie-broken pool majorities, with one of the three distinguishable from zero. That sign depends on the tie-break: one judge reverses under the leaderboard's own rule, and a per-annotator score that needs none leaves one judge with a lean distinguishable from zero. We separate the two regimes. Within MULTIPREF's expert-designated pool the two annotators give different coarse labels on 50.1% of comparisons, evidence against the zero intra-group variability its routing model assumes and distinct from the between-pool contrast its authors left for future work. Code, results and per-call judge outputs: `https://github.com/anik-jha/whose-gold`.

1 Introduction

Every preference benchmark rests on a hiring decision. Someone chose whether to recruit crowdworkers or domain experts, how many to put on each item, and how to collapse them into a label. That choice is normally reported in an appendix and then never mentioned again, because the field's implicit model is that annotators are noisy instruments measuring a single underlying quantity.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Human-AI Coevolution: Measuring Human-Agent Teams in the Agentic Era.

The dataset we study states this assumption explicitly, for its routing model. MULTIPREF [Miranda et al., 2024] deliberately collects two annotator pools and its Limitations section says: *"One of our key assumptions is that there is no variability in intra-group annotators ... the dataset disambiguates between normal and expert crowdworker annotations. We leave this exploration for future work."* Both pools are Prolific crowdworkers who passed the same qualification test; MULTIPREF designates one of them *expert* on education, a graduate degree where the other pool needed a bachelor's, with prompts routed by academic domain, and we use its own labels for the two groups throughout rather than the stronger word "domain expert". We therefore do not read the gaps below as a crowd-versus-domain-expert gap.

Part of that exploration has been done. Zhang et al. [2025] study disagreement in MULTIPREF and HELPSTEER2 directly, build a taxonomy of its causes, show that most disagreement is not annotation error, and release MULTIPREF-Disagreements. Their unit of analysis is the *annotator pair*: they take all permutations of MULTIPREF's four annotators per item, which marginalises over precisely the variable MULTIPREF's own future-work note names. We condition on it instead. That is the difference this paper turns on, and it buys two things Zhang et al. do not report: a like-for-like contrast between two *predefined populations* rating identical items, and what that contrast does downstream to a model ranking and to an LLM judge. Neither corpus-level disagreement nor its causes is our claim; the comparison conditioned on a predefined pool is. Neither corpus assigns annotators to pools at random, so the comparison is observational: it describes how two recruited populations label the same items, not the effect of hiring one instead of the other.

A human-agent team is scored against exactly these labels. When a team's output is graded, or a monitor is validated against "human agreement", the gold standard is whichever pool was hired; if that choice moves half the item-level decisions, it moves the measured competence of the team as much as the team does. That is the sense in which this is a question about human-AI coevolution rather than about dataset hygiene.

Our contribution is a measurement and a distinction:

1. **Item-level divergence between predefined pools is large.** Two pools annotating the same items reverse the winner on 9.2% (MULTIPREF, on the $2{,}885$ items needing no tie-breaking convention) and 8.5% (MT-BENCH, likewise), and change the majority label on 23.6% and 30.5%. On MULTIPREF the two expert-designated annotators on a single item disagree with each other on the coarse A/Tie/B scale 50.1% of the time.
2. **None of it reaches *this* leaderboard, and that is a statement about spacing.** Ranking the same models under expert-gold and normal-gold labels gives $\tau = 1.00$ on both corpora, with no model displaced. But an item-level bootstrap of the same corpus displaces a model in 28% of resamples, so the observed zero held here but fails in more than a quarter of them, not a demonstration that aggregation absorbs the pool. The 28% describes this corpus under this resampling, not a general probability of leaderboard instability (§4).
3. **The distinction that follows.** These two six-model leaderboards keep their ordering despite the item-level divergence; reward-model training, active-annotation routing, and LLM-judge validation all consume *per-item* labels and get no such averaging. We measure one of these and *conjecture* the others: only the judge is tested here.
4. **One such consumer, measured, and how its measurement depends on aggregation.** Three LLM judges, two from one vendor and one from another, all lean toward the normal pool when scored against tie-broken pool majorities. Reweighted from our stratified sample back to the corpus the lean is -2.2, -3.5 and -1.8pp, and one of the three has an agreement CI excluding zero; on the sample as drawn all three are larger. That shared sign depends on the tie-break (§5). "Our judge agrees with humans $X\%$ of the time" is therefore partly a statement about the hiring decision and partly about how split pools were collapsed.

2 Setup

Corpora. MULTIPREF [Miranda et al., 2024] contains $10{,}461$ pairwise comparisons over six models, each annotated twice by its *normal* pool and twice by its *expert-designated* pool on a five-point preference scale. Both pools are Prolific crowdworkers; the designation rests on education, not on a test of domain competence, and we carry MULTIPREF's own two labels rather than asserting a

Table 1: Headline numbers and the evidence behind each. *Bootstrap*: frequencies under prompt resampling of this corpus. *Primary*: designated after the pre-registered sweep. *Exploratory*: added after review. *: 95% interval excludes zero.

Finding	Number	Evidence	Where
Label changes / reversals	23.6% / 9.2% MULTIPREF; 30.5% / 8.5% MT-BENCH	Measured	Table 2
Model ordering, both boards	$\tau = 1.00$, $0/6$ displaced	Measured	§3
Displacement under resampling	27.9% of resamples; 22.6% at the 0.8pp pair	Bootstrap	§4
Boards with $K > 6$ models	not measured	Illustration	Table 6
Judge lean, original tie-break	-2.2, -3.5^*, -1.8pp	Primary	Table 4
Judge lean, per annotator	-1.6, -2.5^*, -0.6pp	Exploratory	Table 5

stronger contrast than the dataset supports. MT-BENCH human judgments [Zheng et al., 2023] contains 3,355 human votes; 292 item cells are rated by both the benchmark's own authors (`author_*`) and recruited expert annotators (`expert_*`), giving an independent pool contrast over a different task and a different six-model set. The two contrasts differ in kind, so agreement between them supports sensitivity to annotator source rather than a shared expert-versus-normal mechanism.

We attempted two further corpora and report the failures, because they explain why this contrast in particular went unmeasured. Both retain far more annotator information than we need; what neither supplies is two *predefined populations* rating the same items. PRISM [Kirk et al., 2024] keeps participant identity and detailed sociodemographic profiles, and studies how feedback varies across participants, but assigns one participant per conversation; different participants rate an identical prompt-response pair only where one happens to recur [Kirk et al., 2024, App. S.5], so there is no systematic fixed-item pool contrast. HELPSTEER2 keeps several annotators' ratings per response, from which Zhang et al. [2025] derive preferences, and HELPSTEER2-PREFERENCE adds pairwise preferences from three to five annotators per item [Wang et al., 2024a,b], but their annotators come from one vendor's pool with no predefined population split to condition on. The scarce ingredient is not per-annotator labels; it is a pre-assigned population label attached to them.

Aggregation. For each pool we take the majority label per item and score a model by its mean credit over the comparisons it appears in. When a pool is split we resolve it by the rule reported in the first row of Table 3, A over B over Tie, but the primary analysis avoids the question entirely (see below). The judge arm of §5 used a different rule, A over Tie over B, for its sampling strata and its reference labels. Rank comparisons use Kendall τ with a *paired cluster* bootstrap: each resample draws the same independent units for both label sources, prompts for MULTIPREF and questions for MT-BENCH, keeping every comparison row of a unit together, since the two rankings differ only in whose labels are used and both corpora reuse an item across model pairs.

Pre-registration. Decision rules were kept in a versioned plan; what preceded each run is author-attested, not third-party registered. The plan's first committed version, which predates the leaderboard results, held a τ-only rule from an earlier team-level design: if $\tau > 0.9$, report the null and do not enlarge the sweep. The conjunctive form adding a CI lower bound > 0.9 first appears in the commit reporting the result, so its timing is unverifiable. The observed τ meets the first rule but, given its wide interval, not the conjunction; either way we report the null and did not enlarge the model set.

Tie-breaking is not a detail here, so we avoid it. Each pool has two annotators and the two experts on an item give different coarse labels on 50.1% of items, so a "pool majority" is decided by convention on roughly half the corpus, and the corpus is skewed 1.78:1 toward B. Our primary analysis therefore consults no convention: we restrict to the 2,885 items (27.6%) where *both* pools are internally unanimous. That conditions on the items both pools found easy, so 9.2% is not an unbiased estimate of the corpus reversal rate and it is not a bound on it in either direction: across the five conventions of Table 3 the rate runs from 2.5% to 29.8%, straddling it. What 9.2% is, and what no other row in that table is, is a number no tie-breaking rule of ours had a hand in. Table 3 gives the sensitivity. Our leaderboard ran a rule ranking A over B over Tie. Four of five conventions, including the one we ran, leave the ranking at $\tau = 1.0000$: under our own rule, 29.8% of items reverse and the leaderboard still does not move. One rule (A over Tie over B), which our leaderboard did not use but our judge arm did, does break it ($\tau = 0.467$); its mirror does not, so the ordering result depends on the aggregation convention.

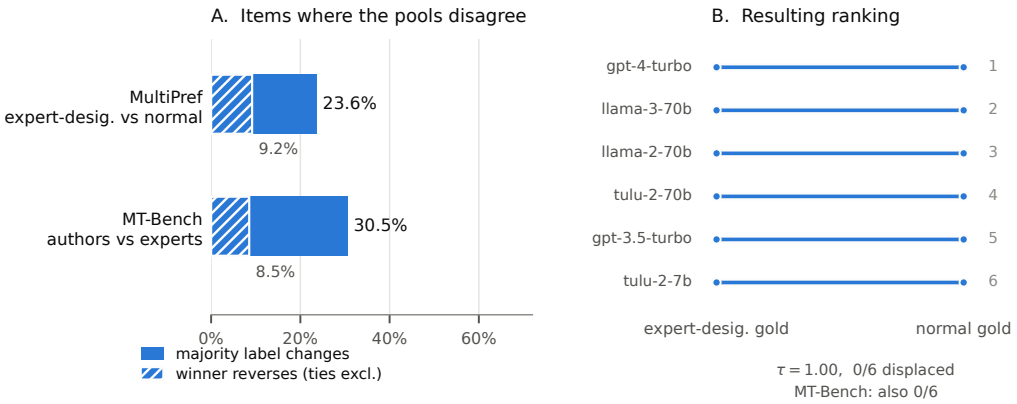


Figure 1: Two pools annotating the same items disagree substantially (**A**) and produce identical model rankings (**B**). Hatching marks outright winner reversals, a strict subset of majority-label changes. Panel B shows MULTIPREF; MT-BENCH likewise displaces zero of six. Panel B is a six-model leaderboard, and §4 shows that is doing much of the work: an item-level bootstrap of the same corpus displaces a model in 28% of resamples.

Table 2: The same pattern on two corpora with different pool contrasts. Winner reversal excludes ties on both sides, so it counts only items where both pools name a decisive and opposite winner. MULTIPREF is restricted to items where both pools are internally unanimous, and MT-BENCH likewise, so no tie-breaking convention enters either corpus.

Corpus	Pool contrast	Items	Majority differs	Winner reverses	Kendall τ
MULTIPREF	expert-desig. vs. normal	2,885	23.6%	9.2%	1.0000
MT-BENCH	authors vs. experts	246	30.5%	8.5%	1.0000

3 Divergence at the item level, invariance in aggregate

Table 2 and Figure 1 give the result. On the convention-free MULTIPREF subset, one comparison in eleven has the two pools' majorities naming opposite winners. Where the divergence sits is a convention-dependent question, so we answer it under all of them. On the convention-free subset, bucketing items by the quality gap between the two models compared gives 30.8%, 24.3% and 15.3% majority divergence across tertiles, from the most closely matched pairs to the most widely separated (Pearson $r = -0.154$). Across the five conventions of Table 3 that coefficient runs from -0.220 to $+0.097$ and *changes sign*: it is negative under three rules including the convention-free one, and positive under the other two. We therefore do not claim that divergence concentrates on closely matched pairs, which would be the convenient result here, since it is the adjacent pairs a displacement needs (§4). The tie-breaking rule decides even the sign of that relationship, and one number that survives every convention is enough: divergence never falls below 15.3%, even in the tertile of most widely separated model pairs.

The ordering is unchanged. Win rates move by -2.1 to $+2.8$pp and the rankings they induce are identical: $\tau = 1.0000$, zero of six models displaced, on both corpora. The MULTIPREF bootstrap CI is $[0.867, 1.000]$ and the MT-BENCH CI is $[0.733, 1.000]$; the intervals are wide because six models admit few discordant pairs, so we claim only that no displacement occurs, not that none could.

A 9.2% item-level reversal rate that leaves these orderings unchanged is still a 9.2% reversal rate for any consumer that does not take the mean. But before drawing comfort from the unchanged ordering, it is worth asking how much comfort six models can supply.

4 How much invariance is that, exactly?

A leaderboard can be invariant because aggregation is robust, or because nothing on it was close enough to swap. Those are different claims and a τ of 1.00 does not separate them, so we measure the second directly.

Table 3: Every tie-breaking convention we tried, on all 10,461 MULTIPREF comparisons (the unanimous row is the 2,885-item subset, where no rule is consulted). The first row is the rule our leaderboard actually used. Only the last, the rule ranking A over Tie over B that our judge arm used (§5), moves the ranking; its mirror image does not, so the ordering result depends on the aggregation convention. The last column is the correlation between a pair's model-quality gap and its divergence rate: the convention decides even the sign, and the flat reading belongs to the same rule the τ column isolates.

Tie-break rule	Majority differs	Winner reverses	Kendall τ (displaced)	r(gap)
A > B > Tie (leaderboard)	44.9%	29.8%	1.0000 (0/6)	−0.081
Both pools unanimous (no rule)	23.6%	9.2%	1.0000 (0/6)	−0.154
Force ties to Tie	41.2%	2.5%	1.0000 (0/6)	+0.097
B > Tie > A (mirror)	36.6%	9.4%	1.0000 (0/6)	−0.220
A > Tie > B (judge arm)	49.8%	16.8%	0.4667 (4/6)	+0.030

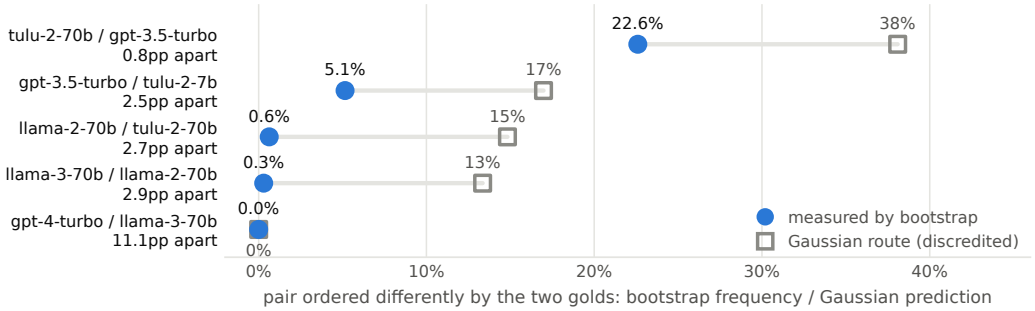


Figure 2: Whole-leaderboard risk is concentrated in one adjacent pair, and the Gaussian route over-predicts every pair with nonzero measured disagreement. Each row is an adjacent pair of the MULTIPREF leaderboard, ordered by the gap between them; the filled circle is the share of bootstrap resamples in which the two golds order that pair differently, the open square is what the Gaussian approximation of §4 predicts for the same pair. The approximation over-predicts all four nonzero pairs, which is why the section reports it as failing rather than as a result.

The perturbation. Switching pools moves each model's win rate: across our six models the normal-minus-expert-designated shift ranges from −2.1pp to +2.8pp, with SD = 1.9pp across models.

Our own leaderboard was lucky, and we measure how lucky rather than model it. Its adjacent gaps are 11.1, 2.9, 2.7, 0.8 and 2.5pp. MULTIPREF reuses each prompt for two comparisons against different model pairs, so the resampling unit is the *prompt* and a whole cluster travels together; 98.2% of rows sit in a doubled prompt, and resampling rows instead moves the rates below by less than a point. Resampling and recomputing both leaderboards puts the whole-leaderboard displacement rate at 27.9%, and the closest pair, 0.8pp apart, is where almost all of it lives: the two golds order that pair differently in 22.6% of resamples, against 0.3–5.2% for every other adjacent pair. The observed zero is the modal outcome, not a reliable one.

A Gaussian route is available and we report it only to say that it does not work here. Treating the two models' shifts as independent draws from the cross-model SD gives a pairwise SD of 2.6pp and a 38% swap probability for the 0.8pp pair, against the 22.6% measured; for the three mid-sized gaps it predicts 13–17% against 0.3–5.2%. The error is in the input, not the algebra: 1.9pp is the dispersion of six models' realised shifts, whereas the quantity a swap depends on is the sampling SD of one *pair's* shift difference, which the bootstrap puts at 1.3pp. Substituting that recovers the bootstrap (32% against 27.9%). The 2.6pp input is therefore too large, and Table 6 is computed from it.

Appendix A gives an illustrative simulation of larger boards; it is not validated, and no claim in the paper rests on it.

This changes what the aggregate null licenses. It is not that leaderboards are robust to who annotates. It is that *this* six-model leaderboard, with one 11pp gap in it and one 0.8pp gap that came up heads, survived, and that the item bootstrap on the very same corpus fails to survive in 28% of resamples.

That much is measured. Whether the reassurance also fails at larger K is the extrapolation of Appendix A, and it needs a corpus with more models to settle. On a 20-model board we would still stand behind the item-level finding but not its exact rates, which describe these items, and behind neither $\tau = 1.00$ nor the 28%, both specific to this board's spacing.

5 What inherits the divergence

A widely deployed per-item consumer is the LLM judge, validated by statements of the form *"our judge agrees with humans $X\%$ of the time."* If a judge tracks one pool more than the other, X is partly a statement about the hiring decision. Neither pool is ground truth here: we measure which one a judge tracks, not which one is right.

We judged a stratified sample of MULTIPREF with three local open-weight models, in both presentation orders, under a schema-constrained five-point output identical to the human scale. All three lean toward the normal pool under the three estimators of Table 4, all of which score against pool majorities tie-broken A over Tie over B. That shared sign does not survive every way of collapsing a split pool (*Convention control*, below). How far is also a smaller and more fragile number than our first draft of this section reported, for two reasons we now handle explicitly.

The sample is not the corpus. Power to separate the pools lives almost entirely in *pool-split* items, where the two majorities disagree, so the sweep deliberately drew 60% of its $2{,}500$ items from that stratum against 49.8% in the corpus. Any agreement figure computed on the sample as drawn therefore overstates the corpus effect. We reweight by the inverse sampling probability within each stratum, which is the estimand a reader cares about (what the judge would have done on all $10{,}461$ comparisons), and report that as primary. It shrinks each of the three effects by 17%.

One of three clears the bar at corpus weighting. The pre-registered rule is conjunctive: kill if the agreement difference *and* the erasure difference are both indistinguishable from zero. Reweighted to the corpus, Qwen3.6-27B clears it on agreement (-3.5pp, CI excluding zero) and Gemma-4 clears it on erasure ($+2.4$pp, CI excluding zero), while Qwen3.6-35B-A3B clears neither (-2.2pp, CI $[-4.4, +0.1]$; erasure $+1.8$pp, CI $[-0.2, +3.9]$) and would be killed under its own rule at that weighting. All three point the same way under this tie-break and all three cleared the bar on the sample as drawn, which is the sweep we pre-registered; we report the corpus-weighted numbers as primary anyway, because the sample was chosen for power and not as an estimate of anything. Three judges giving the same sign is weak evidence on its own: a sign test on three would not reach significance. Holm-adjusting the six tests this bar involves, three judges by two metrics, leaves only Qwen3.6-27B's agreement difference significant (adjusted $p = 0.016$, from the prompt-level resampling below). Intervals outside this six-test family are exploratory. Under this tie-break the summary is a consistent direction, a corpus-level magnitude near 2–3.5pp, and one judge that carries it individually.

On pool-split items, where the stratification does not bias a within-stratum share, Qwen3.6-35B-A3B matches the normal pool on 46.3%, the expert-designated pool on 34.0% and neither on 19.7%; Qwen3.6-27B gives 40.6%, 32.7% and 26.6%; Gemma-4 gives 39.5%, 33.3% and 27.2%. Among the items where the judge matched *either* pool the normal-pool share is 57.6%, 55.4% and 54.3%.

Convention control. The tie-break sets a pool's majority on roughly half the corpus. To test whether the lean comes from that rule, or from the label marginals it induces meeting a judge's own A/B response bias, we rescored the same saved calls under every rule of Table 3, keeping the original strata and weights (Table 5). The sign is rule-dependent: under the leaderboard's rule, and under B over Tie over A, Gemma-4's lean becomes $+0.5$pp. On the 214 sampled items where both pools are internally unanimous and disagree, the judges side with the expert-designated pool 87, 83 and 90 times and with the normal pool 83, 83 and 79 times. A per-annotator score, the judge's mean agreement with a pool's two annotators, needs no tie-break and leaves a lean distinguishable from zero only for Qwen3.6-27B (-2.5pp, Holm-adjusted $p = 0.010$ across the three judges), whose original-rule lean also exceeds what its label marginals predict; about half of Gemma-4's is what its marginals predict. These intervals resample whole prompts (Appendix B); for the original rule this leaves every corpus-weighted agreement and erasure interval on the same side of zero as before.

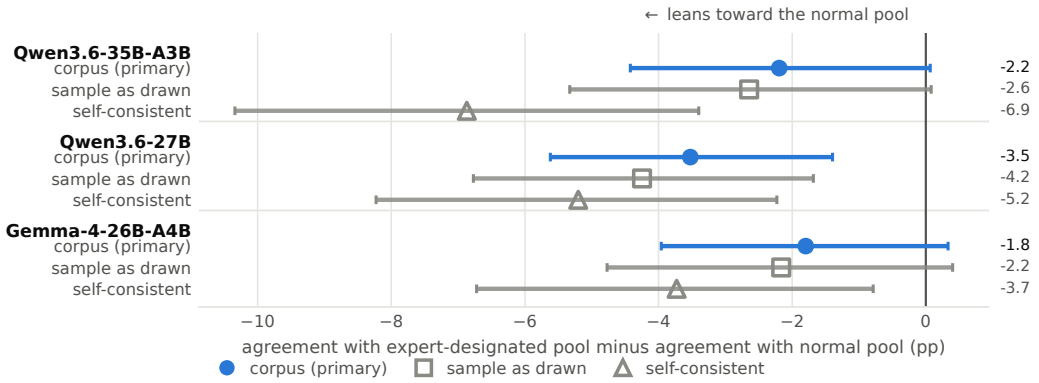


Figure 3: Every judge leans toward the normal pool under all three estimators shown, which share one tie-break (others: Table 5). How far depends on which estimator is designated primary. Points are the difference in agreement with each pool, bars are 95% bootstrap CIs, and the vertical line is no lean. Under the corpus-reweighted estimator the paper treats as primary, one interval of three excludes zero. The self-consistent subset sits systematically further from zero, which is the artifact §5 withdraws rather than a stronger result.

Table 4: LLM-judge alignment with each annotator pool on MULTIPREF. Negative means the judge agrees more with the *normal* pool. **Corpus** is the primary estimator: all items, midpoint order-combination, reweighted by inverse sampling probability from the 60% pool-split sample to the corpus's 49.8%; CIs are a stratified paired bootstrap ($B = 5000$). **Sample** is the same estimator on the sample as drawn, and **Self-consistent** restricts to items where the judge gives the same direction in both presentation orders, a pre-registered restriction that removes uninformative items but also selects a different item set per judge, which is why it is shown last rather than first. Every judge is negative under all three, which share the A over Tie over B tie-break (Table 5). * marks a CI excluding zero. On the corpus estimator the erasure leg of the conjunctive bar is $+1.8$pp $[-0.2, +3.9]$, $+2.0$pp $[-0.0, +4.0]$ and $+2.4$pp $[+0.3, +4.5]^*$ in table order.

Judge	Corpus (primary)	Sample	Self-consistent
Qwen3.6-35B-A3B (Qwen)	-0.022 $[-0.044, +0.001]$	-0.026 $[-0.053, +0.001]$	-0.069 $[-0.103, -0.034]^*$
Qwen3.6-27B (Qwen)	-0.035 $[-0.056, -0.014]^*$	-0.042 $[-0.068, -0.017]^*$	-0.052 $[-0.082, -0.022]^*$
Gemma-4-26B-A4B (Google)	-0.018 $[-0.040, +0.003]$	-0.022 $[-0.048, +0.004]$	-0.037 $[-0.067, -0.008]^*$

These checks were added after review and are exploratory; they show that the published direction partly reflects the aggregation rule, not what mechanism produces it.

We flag two honest complications, and a claim we are withdrawing on the strength of the second. First, an initial run of the second model at half this sample size did *not* clear the pre-registered replication bar (-0.8pp, CI $[-4.4, +2.8]$, against -4.2pp at full power under that same estimator); the effect appeared only once power was matched, and we report that rather than presenting the matched run as the first attempt.

Second, we had explained the effect as attenuation by label noise, which predicts that a cleaner judge shows a *larger* effect. On the self-consistent subset the data ran exactly the other way: order-swap direction reversal falls $44.6\% \to 24.5\% \to 23.7\%$ while the lean falls $6.9 \to 5.2 \to 3.7$pp. An earlier draft of this paper offered that monotone pattern as the sharpest clue toward a replacement mechanism. It does not survive the primary estimator. Reweighted to the corpus the three leans are 2.2, 3.5 and 1.8pp against those same inconsistency rates: not monotone, and the noisiest judge is in the middle rather than at the top. The pattern was an artefact of the subset itself. Conditioning on self-consistency keeps a different fraction of items for each judge (55% for the most position-sensitive model against 76% for the least), and the items a judge is confident about are items where the pool lean is larger, so the estimator rewards inconsistency with a bigger number. The attenuation explanation stays withdrawn, its replacement is withdrawn too, and the mechanism is simply open.

Table 5: Convention control: corpus-weighted agreement with the expert-designated pool minus agreement with the normal pool, in pp, from the same saved calls, original strata and weights. The first four columns tie-break each pool's majority differently; *Per annotator* needs no tie-break; *Excess* is the original-rule difference minus its expectation under independent labels within strata. * marks an unadjusted prompt-resampled 95% interval excluding zero (Appendix B). Exploratory.

Judge	A>Tie>B	A>B>Tie	B>Tie>A	Force Tie	Per annotator	Excess
Qwen3.6-35B-A3B	-2.2	-1.8	-1.0	-2.7^*	-1.6	-2.0
Qwen3.6-27B	-3.5^*	-2.2^*	-1.4	-3.3^*	-2.5^*	-2.7^*
Gemma-4-26B-A4B	-1.8	$+0.5$	$+0.5$	-3.2^*	-0.6	-0.9

Two further measurement choices move the judge more than the pool does. Both are by-products of the judge arm rather than targets of it, and both are far larger than the 1.8–3.5pp corpus-weighted pool effect the section is about, which is the reason to record them. First, all three judges are strongly position sensitive, reversing direction under order swap on 44.6%, 24.5% and 23.7% of items, consistent with published position-bias measurements [Mazur, 2026] and, importantly, differing by a factor of 1.8 between the two Qwen models despite being the same family. Second, holding task content, model, items, presentation orders, temperature and seed fixed while changing the output-format instruction and its decoding constraint together (a JSON object versus a bare constrained token), the same judge agrees with itself on only 47.2% of items ($n = 500$), with a systematic shift in one direction (146 items move A→B, 3 move B→A). Judge numbers are conditional on output format, a sensitivity of the kind Bellibatlu et al. [2026] finds largely unexamined. Appendix C reports a partial localisation of this effect.

6 What to do instead

Report the pool and the aggregation rule. An LLM-judge agreement figure without the annotator pool that defines its reference labels, and the rule that collapses a split pool into one label, is underspecified by roughly the size of the effect being reported.

Do not infer per-item reliability from leaderboard stability. These are different quantities and our data separates them by a wide margin. Nor should a leaderboard's stability be read off the leaderboard itself: at six models the $\tau = 1.00$ is partly a statement about how far apart the models were, and the item-level bootstrap in §4 is the cheap check that says so.

Counterbalance order, always. At 23.7–44.6% direction reversal, a single-order judge measurement on this task is unstable on about a quarter to nearly half of its items.

Weight the sample you drew, and report the unrestricted estimator. Stratifying a judge sweep toward the items that carry the signal is the right way to spend inference, and it makes every raw agreement rate a statement about the sample rather than the corpus; reweighting cost us 17% of the effect. The larger cost came from elsewhere: restricting to items where the judge is self-consistent makes all three effects roughly double and all three CIs exclude zero, and it keeps a different item set for each judge. Report the unrestricted estimator first and the restriction as a sensitivity analysis, not the other way round.

Assign populations, not just annotators. Per-annotator labels are necessary and not sufficient: PRISM keeps rich participant profiles and HELPSTEER2-PREFERENCE keeps multiple annotators per item, and neither provides systematic predefined-pool assignment: the first because one participant sees each conversation, the second because there is no population split to condition on. A corpus that assigns two named populations to the same items, as MULTIPREF does, is what makes the hiring decision auditable after the fact.

7 Limitations

Both corpora rank only six models. The aggregate invariance is therefore a statement about this leaderboard's spacing, and §4 measures rather than asserts what that costs by bootstrapping the same corpus. The $K > 6$ rows of Table 6 are a different kind of object and we label them as such throughout: they extrapolate from six observed shifts under assumptions (i.i.d. Gaussian shifts

across models, uniform spacing across the observed span) that six models cannot check, and they are not bounds. Their agreement with the $K = 6$ bootstrap is not a validation either: Appendix A withdraws that claim, having found it to be two compensating errors that cancel at $K = 6$ under uniform spacing and nowhere else. A corpus with 20 or more models rated by two predefined pools would settle the question directly and is the single experiment that would most improve this paper. The MT-BENCH contrast rests on 246 convention-free cells of 292 doubly-rated ones. The judge arm spans two vendors: LAGUNA S 2.1 does not load in our stack, and a substitute prefill path reproduced the generation path on 94.6% of items against a pre-registered 95% bar, so we ran a third judge from a different pretraining lineage instead, at the same Q8_0 precision as the other two. That third judge is not a controlled vendor swap, because architecture, pretraining corpus and post-training all change with the lineage. It is a replication across three unlike systems rather than an isolation of any one factor, and unlike systems may be less independent than a vendor count suggests: in a different setting, 24 open-weight monitors scoring backdoored code, a monitor's agreement with the rest of its pool tracks its own detection skill (Spearman 0.84) rather than its pretraining lineage ($+0.064$, $p = 0.18$) [Jha, 2026]. Under the original tie-break the normal-pool lean holds in sign across all three, but at corpus weighting only one of the three has an agreement CI excluding zero, a consistent sign across three judges is not itself significant, and for one judge the sign depends on the rule. Three models from two vendors is not a claim about judges in general. A fourth judge or a reward-model probe would widen the sample but not settle which aggregation rule defines the reference label, which the convention control addresses. We report the null, the failed under-powered run, and the withdrawn explanation alongside the positive results deliberately: the paper's claim is about measurement validity, and it would be self-undermining to present it selectively. All local judge inference ran on a single on-premises NVIDIA GB10 (Grace Blackwell) workstation serving Q8_0-quantised GGUF checkpoints; the three judge sweeps (2,500 comparisons $\times$ 2 presentation orders each) took 49–122 minutes wall-clock per model.

8 Related work

The closest work. Zhang et al. [2025] is the paper this one has to be measured against. They analyse disagreement in MULTIPREF and HELPSTEER2, build a ten-category taxonomy of its causes, establish that most disagreement reflects genuine differences in taste rather than annotation error, release MULTIPREF-Disagreements, and show that Bradley–Terry reward models and LLM-as-Judge evaluation both fail to represent divergent annotators. Everything in this paper about *whether* MULTIPREF annotators disagree is downstream of that result, not a discovery of ours, and our 50.1% within-pool figure is a recomputation in their territory. Two things separate the papers. Their unit is the annotator pair, taken over all permutations of the four annotators on an item, which averages across the normal/expert designation; ours is the *pool*, held fixed by MULTIPREF's own assignment, which is the contrast its authors named as future work and the one Zhang et al. marginalise over. And they do not follow disagreement into a model ranking: there is no leaderboard or Kendall τ analysis in their paper, whose win/loss/tie rates compare response types under an LLM judge, so the question of whether pool choice reaches the thing benchmarks actually report is open after it and answered here only for two six-model boards.

The rest. Perspectivist evaluation argues that annotator disagreement is signal rather than noise [Xu and Jurgens, 2026]. STABLEVAL [Bonagiri et al., 2026] is the nearest antecedent on the aggregate side: it models annotator confusion and reports that majority vote yields ranking *instability* under annotator heterogeneity, where we find a stable ranking under a specific, pre-assigned heterogeneity. Kadekodi et al. [2025] take the other route open to an aggregator, abstaining from comparisons where fewer than a chosen fraction of individuals agree and returning a partial order rather than arbitrating dissent; that is a safer rule for a fixed set of voters, where we ask the prior question of which population supplies them. These results are compatible: our finding concerns two particular recruited populations and six-model leaderboards, and our own bootstrap displaces a model in 28% of resamples. But a reader should treat STABLEVAL's finding, not ours, as the general statement about majority vote. Work on LLM-judge bias documents position effects [Mazur, 2026] and instability under semantically equivalent prompt rephrasings [Bellibatlu et al., 2026]; our position numbers replicate that line rather than extend it, and our format numbers are reported alongside them as controls on the pool measurement.

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A Larger boards: an illustration, not a measurement

What that would mean for a larger board, if the perturbation carried over. The question a reader asks next is what happens at the leaderboard sizes the field actually publishes, and we can only answer it by extrapolation, so we mark it as one. Table 6 holds the measured shift distribution fixed, assumes those shifts are i.i.d. Gaussian across models, spreads K models uniformly over the observed win-rate span, and simulates. Displacement probability rises from 0.29 at $K = 6$ to 0.92 at $K = 10$ and to 1.00 by $K = 20$.

Three cautions, because the arithmetic is much more confident than the evidence under it. The shift distribution is estimated from *six* models, so its SD is itself poorly determined. Independence across models is assumed and, on the evidence above, wrong: the measured pairwise SD is half what independence implies. And uniform spacing over one corpus's observed span is a stipulation about a hypothetical leaderboard, not a fact about any real one. An earlier draft of this paper offered the $K = 6$ row as a validation, since the simulation's 0.293 sits next to the item bootstrap's 0.279. That agreement is a coincidence of two compensating errors and we withdraw the claim: run the identical simulation on the *observed* gaps rather than uniform ones and it returns 0.70 against the same measured 0.28. Uniform spacing is more forgiving than this board's real gap structure, and the 2.6pp input is roughly twice the measured pairwise SD; the two cancel at $K = 6$ uniform and nowhere else. Table 6 is therefore an illustration of what a *mis-calibrated* perturbation would imply. Its two errors push in opposite directions: the inflated 2.6pp input upward, uniform spacing downward against this board's real gaps. We therefore do not claim to know the net direction of the $K > 6$ rows, and no row of it is checked.

Table 6: **An illustrative scenario, not a measurement.** Probability that changing the annotator pool displaces at least one model, holding the measured per-model shift (SD $= 1.9$pp) fixed, assuming those shifts are i.i.d. Gaussian across models, and spreading K models uniformly across the observed win-rate span. Probabilities are Monte Carlo over per-model shifts (2×10^5 draws), so adjacent pairs correctly share a model; the closed form $1 - (1 - p)^{K-1}$ used in an earlier draft treats the $K - 1$ swap events as independent, which they are not, and understates $K = 10$ by 5pp. **No column here is validated.** The $K = 6$ row sits close to the item bootstrap's 0.279 only because uniform spacing happens to offset an inflated shift SD; on the observed gaps the same simulation returns 0.70. Read the column as an illustration, not as a bound.

Models on the leaderboard	6	10	20	30	50
Gap between adjacent models	4.0pp	2.2pp	1.1pp	0.7pp	0.4pp
P(at least one displaced)	0.29	0.92	1.00	1.00	1.00

B Reproduction details

Code, results files, per-call judge outputs, a label-only MULTIPREF derivative (identifiers and A/Tie/B labels, no prompt or response text) and excerpts of the decision-rule ledger are released at `https://github.com/anik-jha/whose-gold`. The leaderboard analyses rerun from the public corpora, the convention control replays from the saved judge calls without model inference, and a verifier script checks 106 of the reported numbers against their results files. Regenerating the judge calls requires the three checkpoints named below and a `llama.cpp` server.

The stratified sample is $2{,}500$ items drawn with `numpy` PCG64 seed 17, 60% from the pool-split stratum, from the $10{,}461$ MULTIPREF comparisons carrying both pools; the same draw is reused for every judge. The reweighting in Table 4 of §5 gives each scored item the weight $N_{\text{stratum}}/n_{\text{stratum}}$, where n is that judge's *scored* count in the stratum rather than the drawn count, since a handful of items fail to parse ($5210/1498$ and $5251/998$ for the first two judges, $5210/1497$ and $5251/998$ for the third). Judges are served by `llama.cpp` at Q8_0 from `Qwen_Qwen3.6-35B-A3B-Q8_0.gguf`, `Qwen_Qwen3.6-27B-Q8_0.gguf` and `google__gemma-4-26b-a4b-it-Q8_0.gguf`, temperature 0, server seed 1234, with output constrained to the JSON schema `{"p": <"1".."5">}` whose five codes are the five points of the human preference scale; the format-sensitivity check replaces it with the grammar `root ::= [1-5]`. Every item is issued in both presentation orders and combined at the midpoint. Bootstraps are $B = 5000$ (corpus and sample-as-drawn judge estimates, seed 0), $B = 10{,}000$ (self-consistent judge estimates, seed 17), $B = 2000$ (leaderboard τ intervals, seed 0) and $B = 4000$ (displacement analysis, seed 0); the convention control resamples whole prompts, since 604 of each judge's scored rows share a prompt: each prompt draws one mean-one exponential weight, shared across strata and judges, and each stratum is renormalized to its corpus mass ($5{,}000$ replicates, seed 20261002). The illustrative K-model simulation uses 2×10^5 draws.

For the convention control we report percentile 95% intervals and approximate two-sided centered-multiplier p-values, $(1+\#\{b : |\hat{\theta}_b - \hat{\theta}| \geq |\hat{\theta}|\})/(B+1)$. This is an exploratory sensitivity analysis for a superpopulation of independent prompts, conditional on the original stratum masses; it is not exact inference under the finite-population sampling design, and dependence through repeated annotators is not modelled. The stars in Table 5 use these intervals without adjustment. Holm adjustment is applied separately to two exploratory families: the six original-rule agreement and erasure tests of §5, and the three per-annotator tests. The per-annotator differences and intervals, in pp and table order, are -1.6 $[-3.23, +0.17]$, -2.5 $[-4.02, -0.81]$ and -0.6 $[-2.27, +1.05]$.

C Judge prompt

The system prompt below is the one the sweep served, verbatim. The paper shows that judge output is highly sensitive to the response format (§5), which makes the exact string load-bearing rather than incidental, so it is reproduced here rather than left to the code release. Decoding is constrained to the JSON schema `{"p": <"1".."5">}`. The format-sensitivity check of §5 replaces both the constraint, with the grammar `root ::= [1-5]`, and the final sentence of the system prompt, which becomes `Reply with exactly one digit and nothing else:` in place of the JSON instruc-

tion. Those two change together, which is why §5 does not attribute the effect to the scaffolding alone. Everything else is identical.

```
You are evaluating two candidate responses to a user's message. Judge which
response is better overall, considering helpfulness, truthfulness, and
harmlessness. Reply with JSON {"p": "<code>"} using this scale:
  1 = A-is-clearly-better
  2 = A-is-slightly-better
  3 = Tie
  4 = B-is-slightly-better
  5 = B-is-clearly-better
```

The user message is templated as:

```
User's message:
{prompt}

--- Response A ---
{a}

--- Response B ---
{b}

Which response is better overall?
```

Each field is clipped from the middle to 6,000 characters, keeping both ends. Every item is issued in both presentation orders; in order `ba` the two completions are swapped before templating.

The format effect of §5 can be partly localised: re-scoring the same items with the same model at bf16 (rather than the Q8_0 quantisation the sweep served) by *prefilling* the JSON scaffolding and reading the next-token distribution recovers 94.6% agreement with the generated labels, against 47.2% for a bare digit. That check is not clean: it changes precision (bf16 against Q8_0) and decoding path (prefill against generation) at the same time, so the 5.4% residual absorbs both and we cannot separate them. The prefill check thus disagrees with the generated labels on 5.4% of items, against 52.8% for the JSON-versus-digit condition. That condition is itself compound: it changes the system's output-format instruction as well as the decoding constraint, so this audit cannot separate instruction wording from JSON scaffolding. A clean attribution needs the constraint varied at fixed instruction wording, fixed precision and fixed decoding path, which we did not run.

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